Zero-Trust Network Architecture Framework

A comprehensive zero-trust segmentation framework for enterprise identity platforms.

#Overview

This framework enforces least-privilege access across microservice architectures using mutual TLS, SPIFFE/SPIRE workload identity, and dynamic policy enforcement. Designed for enterprise environments with 50+ microservices requiring strict identity verification for every network request.

#Architecture

The system consists of three core layers: an Identity Provider (SPIFFE/SPIRE) that issues short-lived SVIDs to workloads, a Policy Engine that evaluates access requests against fine-grained OPA policies, and an Envoy-based Data Plane that enforces mTLS and authorization at the network level.

#Tech Stack

- Language: Go, Python

#Installation

git clone https://github.com/danalevy/zero-trust-framework

#Usage

Configure workload identities and access policies in YAML. The framework automatically provisions SPIFFE identities and enforces mTLS between services.

#Key Features

- Mutual TLS enforcement across all service-to-service communication

#Performance

The framework adds less than 2ms latency per request for mTLS handshake and policy evaluation. Tested with 10,000 concurrent connections across 50 microservices.

#Contributing

Contributions welcome. Please follow security review guidelines in CONTRIBUTING.md. All PRs require security-focused code review.

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